Analytical Cumulative examination

16th January 1999

The excerpt below is from the www description of an instrument on the Stardust spacecraft that is to be launched in three weeks. The mission will fly a spacecraft through the tail of a comet and return collected dust particles, and particles of interstellar dust, to Earth. It is the first sample return mission to be launched into space since the Apollo lunar program.

The techniques used on the spacecraft for *in situ* measurements are simple and well-known to analytical chemists. Write an essay explaining completely and quantitatively the theoretical background to the instruments and any experimental details you think will have been addressed in developing the instruments for space flight.

Comet and Interstellar Dust Analyzer (CIDA)

This is the same instrument design as flew on Giotto and two Vega spacecraft, obtaining unique data on chemical composition of individual particulates in Comet Halley's coma. It consists of an inlet, a target, an ion extractor, a time-of-flight (TOF) mass spectrometer (MS) and an ion detector. The inlet is a baffled system to prevent sunlight from entering the instrument and raising the background noise in the detector. The target is planned to be a corrugated Ag or other heavy metal material. It will not be necessary to have a moving Ag foil for the Comet Wild flyby, as was done for the higher flux at Halley. In addition, the target area will be increased to 50 cm^2 from the previous 5 cm^2. A light flash which accompanies the initial impact is detected and used to set the zero for the TOF measurement. Electrostatic grids extract positive or negative ions, depending upon which polarity they are commanded to, from the impact microplasma. These ions move down a bent-tube TOF MS, with an electrostatic reflector to focus ions of similar energies onto the ion detector. By measuring arrival time, the mass of the ions is determined. It is expected that at the 6.1 km/s flyby speed, molecular ions as well as atomic ions will be important. The instrument is sensitive at least over the range AMU=1 to 150. Even sub-micron sized particles produce observable signals and compositional profiles.

The use of a transient recorder mode will allow a much superior data set than was possible from the data-constrained links that were available for the previous particle impact analyzer (PIA) flights because it will take all data at the equivalent of Mode 0 data, i.e., the highest time resolution and full peak profiles rather than peak detection. In addition, it will eliminate the criticality of the event detection methods since all signals generated by the ion detector itself will be accepted rather than gated. For the comet flyby, most of this data will be played back slowly over ensuing days or weeks.

The investigator who designed and developed PIA for the Halley missions is a member of our science team and will lead its re-build and enhancements for this advanced application. The German DLR is fully backing the flight of this unit on the STARDUST mission.